ISA104 / AI APPLICATION DESIGN

Human-Centred Design with AI

Week 9

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XJTLU

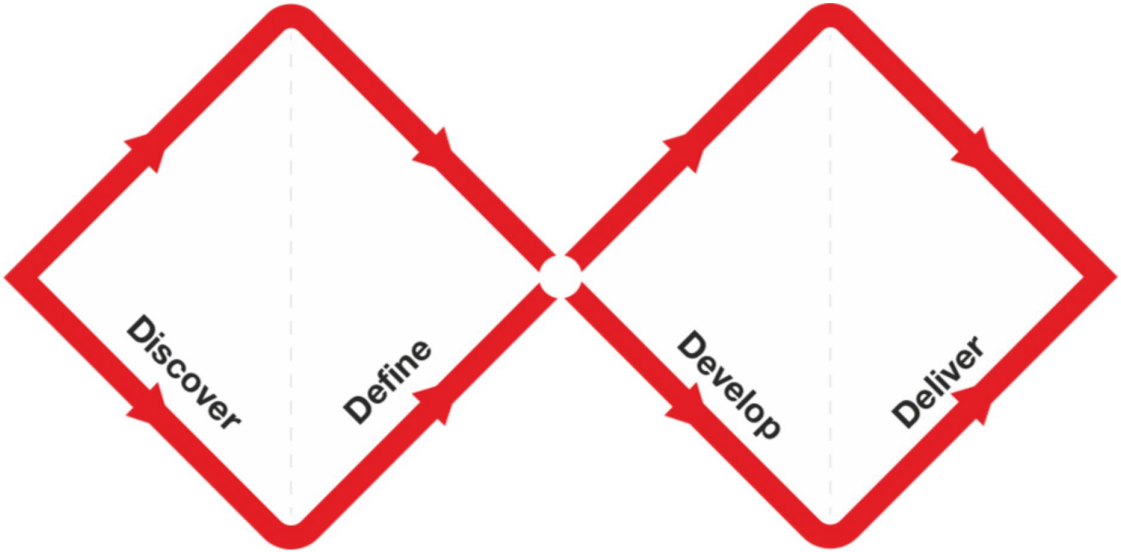
Learning Outcomes

A framework for improving your group Agent

DESIGN PROCESS	Explain the Double Diamond and connect a user need to a design decision.
EVALUATION GOALS	Distinguish utility, usability and other outcomes, then choose suitable evidence.
AI AND YOUR AGENT	Use paper examples to plan AI assistance and the checks people need to make.

The Double Diamond

Discover, Define, Develop, Deliver



Original Design Council diagram, CC BY 4.0.

Explore and focus the problem, explore solutions, then test and improve. Revisit earlier stages.

Design Council, The Double Diamond. Classroom stage mappings are our interpretation.

Stages, Methods and Outputs

The stage defines the purpose; the method provides evidence for the next decision

STAGE	SAMPLE METHODS	USEFUL OUTPUT
Discover	Observation, interview, survey	Evidence about users and context
Define	Coding, affinity mapping, requirements	Focused problem and design requirements
Develop	Brainstorming, sketches, prototypes	Alternative solutions and a chosen design
Deliver	Usability test, experiment, questionnaire	Evaluation evidence and revisions

The stage describes the purpose. A method can support more than one stage.

Design Council, The Double Diamond. Classroom stage mappings are our interpretation.

Discover: Questions and Methods

Choose the method that can answer the question

METHOD	WHAT IT CAN REVEAL	AGENT PROJECT EXAMPLE
Field study	Behaviour in a real setting	Observe use at a busy help desk.
Interview	Experiences, opinions and reasons	Ask about a recent difficult task.
Survey	Self-reported patterns across people	Ask how often a difficulty occurs.

These are proposed Agent investigations. An interview does not require a fixed order.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Interview Types

Preparation and flexibility shape the conversation

STRUCTURED

Prepared questions
Fixed wording and order
Easy to compare answers

SEMI-STRUCTURED

Prepared topics or questions
Follow-up questions allowed
Depth with flexibility

UNSTRUCTURED

A broad topic or purpose
Conversation develops openly
Useful for exploration

A relevant follow-up: “What happened just before you gave up?”

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Individual Interview and Focus Group

Who speaks with whom changes the evidence

INDIVIDUAL INTERVIEW

One participant speaks with a researcher.

Explore a personal experience in detail.

Useful when individual or private concerns matter.

FOCUS GROUP

Participants discuss a topic together.

They respond to one another's experiences.

A moderator guides the discussion and balances voices.

Discussion: Which method could reveal disagreement about your Agent's proposed role?

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Define: From Evidence to Requirements

Turn research material into a focused, evidence-based design problem

STEP	METHOD	USEFUL OUTPUT
Organise evidence	Code and group records	Traceable patterns
Interpret findings	Identify needs and tensions	Evidence-based insights
Frame the problem	Choose the user, task and context	A focused problem
Write requirements	State the support the design must provide	Testable requirements

A requirement should explain the support needed without fixing the interface too early.

Wu (2024), Designing Human-Centered Software, CMU. Classroom synthesis adapted for ISA104.

Develop: Ideas and Prototypes

Explore alternatives before choosing and refining one design

ACTIVITY	METHOD	USEFUL OUTPUT
Generate alternatives	Brainstorming or co-design	Several concepts
Make ideas visible	Sketches or storyboards	Interaction scenarios
Build a version	Wireframe, paper or interactive prototype	A testable prototype
Compare choices	Requirements and design principles	A reasoned selection

More ideas help only when the team compares them with the requirements.

Design Council, The Double Diamond. Classroom method mapping adapted for ISA104.

Interface Design Principles

Reduce memory demand and help people understand, control and correct an Agent

RECOGNITION	Show labelled options and relevant evidence instead of hidden commands.
VISIBLE STATUS	Show whether the Agent is receiving input, working or finished.
USER CONTROL	Let users inspect, accept, dismiss or revise a proposed action.
ERROR RECOVERY	Provide a way to correct input or recover from an unwanted result.

Discussion: Which principle would improve one interaction in your Agent?

Nielsen (1994), 10 Usability Heuristics for User Interface Design, NN/g. Agent examples adapted for ISA104.

Formative and Summative Evaluation

The purpose of the study determines the distinction

FORMATIVE

Find problems so the design can improve.

Example: try a paper prototype and revise the flow.

Evidence supports the next design iteration.

SUMMATIVE

Assess how well a design meets its goals.

Example: compare final task performance or collect final ratings.

Evidence supports an overall assessment.

Questionnaires and observation can support either purpose.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Utility and Usability

Two questions about an application's value to its users

UTILITY

Does the functionality meet the user's needs?

Agent example: supports a revision plan that fits the student's real timetable.

USABILITY

How easy and pleasant are the functions to use?

Agent example: new users can find and change the plan without help.

A useful application needs appropriate functions and a usable way to access them.

Nielsen (2012), Usability 101: Introduction to Usability, NN/g.

Five Usability Components

Nielsen's framework, with possible measures for our Agent

COMPONENT	QUESTION	POSSIBLE EVIDENCE
Learnability	How easy is first use?	Success and help needed on a first task
Efficiency	How fast is use after learning?	Task time and unnecessary steps
Memorability	How easy is returning after a gap?	A return session after a stated interval
Errors	What goes wrong, and can users recover?	Error frequency, severity and recovery
Satisfaction	How pleasant is use?	Post-use ratings and explanations

Choose users, tasks and a context before choosing a measure.

Nielsen (2012), Usability 101: Introduction to Usability, NN/g.

Usability in a Context of Use

ISO 9241-11:2018 focuses on outcomes for specified users and goals

EFFECTIVENESS	How accurately and completely can users achieve the goal?
EFFICIENCY	What resources, such as time and effort, do they use?
SATISFACTION	How well does the experience meet their needs and expectations?

This is another framework. Interpret it separately from Nielsen's five components.

ISO (2018), ISO 9241-11: Usability, Definitions and Concepts.

Evidence for Utility

A classroom checklist for deciding whether functions address real needs

CHECK	QUESTION	EVIDENCE TO SEEK
Need coverage	Which important needs do the functions cover?	Link each function to user evidence
Task and context fit	Can the function support the intended situation?	Try a relevant task in its context
Goal benefit	Does it help achieve a valued outcome?	Assess outputs and perceived usefulness

These are our evaluation questions, rather than a universal three-factor Utility scale.

Nielsen (2012), Usability 101: Introduction to Usability, NN/g.
Davis (1989), Perceived Usefulness and Perceived Ease of Use, MIS Quarterly.

Paper Prototype Evaluation

Classroom exercise: an Agent answer with a visible evidence link

1 USER	Try to find the source behind the answer.
2 FACILITATOR	Present the task and simulate the interface response.
3 OBSERVER	Record confusion, then suggest a revision.

Use feedback to improve the prototype before building more of the system.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

The Think-Aloud Method

Listen to what participants express while they use the interface

PARTICIPANT	Perform the task and explain what you are thinking.
FACILITATOR	Give neutral reminders, such as “Please keep talking.”
RESEARCHER	Record actions, hesitation and comments to investigate interface problems.

The comments are evidence about expressed thinking during the task.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Within and Between Subjects

The participant allocation determines who experiences each condition

WITHIN-SUBJECTS

The same participants experience all conditions.

Each person uses interface A and interface B.

Consider learning, fatigue and carry-over.

BETWEEN-SUBJECTS

Different groups experience different conditions.

One group uses A. Another group uses B.

Reduces carry-over, while people may differ between groups.

Match tasks and procedures. Random assignment can help balance groups.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.

Learning and Carry-Over Effects

Classroom example: order can change the second attempt

ORDER	WHAT COULD HAPPEN	RESPONSE
A then B	Practice with A makes B seem easier.	Include the reverse order.
B then A	Fatigue makes A seem harder.	Balance order across people.
Same task again	Participants remember the answer.	Use comparable tasks and balance materials.

InterLink counterbalanced interface order. Counterbalancing reduces systematic order effects.

Wu (2024), Designing Human-Centered Software, CMU. Teaching examples adapted for ISA104.
Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104

A Goal Needs a Measurement Plan

A classroom example for an Agent that helps students plan revision

ELEMENT	EXAMPLE	WHY IT MATTERS
Users and task	First-year students change a revision plan	Defines the context
Design claim	Visible controls make first use easier	Specifies learnability
Behavioural evidence	First-task success and help requests	Shows actual performance
Subjective evidence	Post-use usability ratings and comments	Shows perceived experience

AI can suggest a plan. Your group must justify the task and the evidence.

Nielsen (2012), Usability 101: Introduction to Usability, NN/g.
ISO 9241-11:2018. Classroom measurement example.

Structured Questionnaires

Choose an instrument for the construct you intend to measure

INSTRUMENT	STRUCTURE	MAIN INTERPRETATION
SUS	10 items, 5-point responses	Overall perceived usability
PU and PEOU	Original final scales: 6 items each	Perceived usefulness and perceived ease of use
USE	30 items across four categories	Usefulness, ease of use, ease of learning, satisfaction

Questionnaire scores describe perceptions. Add behavioural evidence where the claim requires it.

Brooke (1996), SUS. Davis (1989), PU and PEOU.
Lund (2001), Measuring Usability with the USE Questionnaire.

System Usability Scale

Brooke's SUS provides a global subjective assessment after use

STRUCTURE	10 items with positive and negative wording, rated on five response points.
SCORE	A composite score from 0 to 100. It is a scale score rather than a percentage.
INTERPRETATION	Use the overall score. It does not provide five independent component scores.

What additional evidence would you collect for actual efficiency or memorability?

Brooke (1996), SUS: A "Quick and Dirty" Usability Scale.

Perceived Usefulness and Ease of Use

Davis's scales distinguish perceived task benefit from perceived effort

PERCEIVED USEFULNESS (PU)

Original final scale: six items about perceived work-related benefit.

Classroom paraphrase: "This Agent would help me complete the task more effectively."

PERCEIVED EASE OF USE (PEOU)

Original final scale: six items about perceived ease and effort.

Classroom paraphrase: "Learning to use this Agent would be easy for me."

PU items concern one construct. They are not six independent Utility dimensions.

Davis (1989), Perceived Usefulness, Perceived Ease of Use, and User Acceptance of IT, MISQ.

USE and Questionnaire Adaptation

The original instrument and the version used in VisTellAR

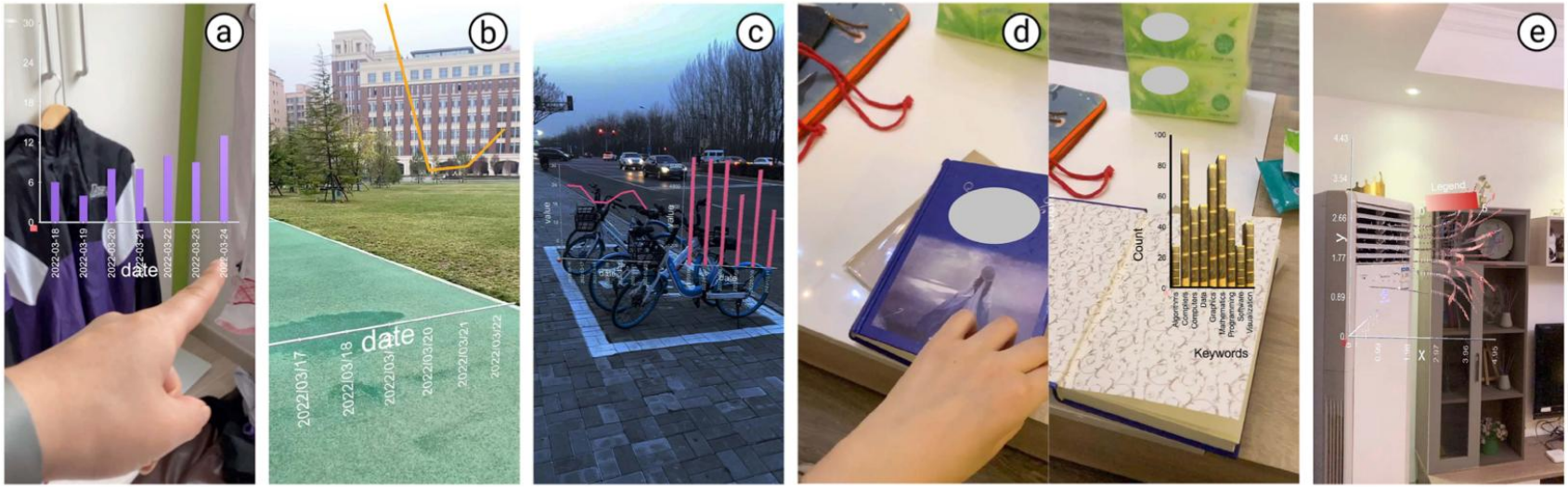
FEATURE	ORIGINAL USE	VISTELLAR ADAPTATION
Length	30 items	10 items
Response scale	7-point agreement	5-point agreement
Content	Four categories	Selected and adapted effectiveness and usability items
Reporting	Name the instrument and its source	Explain the adaptation and interpretation

A citation to a scale does not make an adapted questionnaire identical to its original.

Lund (2001), USE. Tong et al. (2025), VisTellAR, §5.2 and Fig. 8.

Research Example: VisTellAR

Mobile AR authoring for short videos with data visualisations



Original Figure 7.

User need: express an everyday story with relevant data while filming.

Tong et al. (2025), VisTellAR, IEEE TVCG 31(3). DOI: 10.1109/TVCG.2024.3372104

DEMO VIDEO

VisTellAR's Design Process

The paper documents needs, requirements, design and evaluation

OUR STAGE MAPPING	DOCUMENTED WORK	PAPER SECTION
Discover	10-person formative study using three design probes	Section 3
Define	Four main design requirements	Section 3
Develop	Mobile AR authoring and later editing	Section 4
Deliver	Two-day take-home study with 12 creators	Sections 5–6

Double Diamond is our classroom mapping of the documented work.

Tong et al. (2025), VisTellAR, IEEE TVCG 31(3). DOI: 10.1109/TVCG.2024.3372104

VisTellAR: Requirements and Choices

Examples of how a requirement can lead to an interface feature

REQUIREMENT	REPORTED DESIGN CHOICE	DISCUSSION
Connect data with the physical setting	Anchor visualisations to detected objects or planes	What context must the system understand?
Configure visualisations in situ	Edit data and appearance before recording	Which user need does this support?
Support later adjustment	Edit position, scale and timing after recording	What should remain under user control?

A feature needs a reason connected to a user requirement.

Tong et al. (2025), VisTellAR, IEEE TVCG 31(3). DOI: 10.1109/TVCG.2024.3372104

VisTellAR: What Do the Items Measure?

Compare the original figure's labels with the construct behind each item

TASK SUPPORT

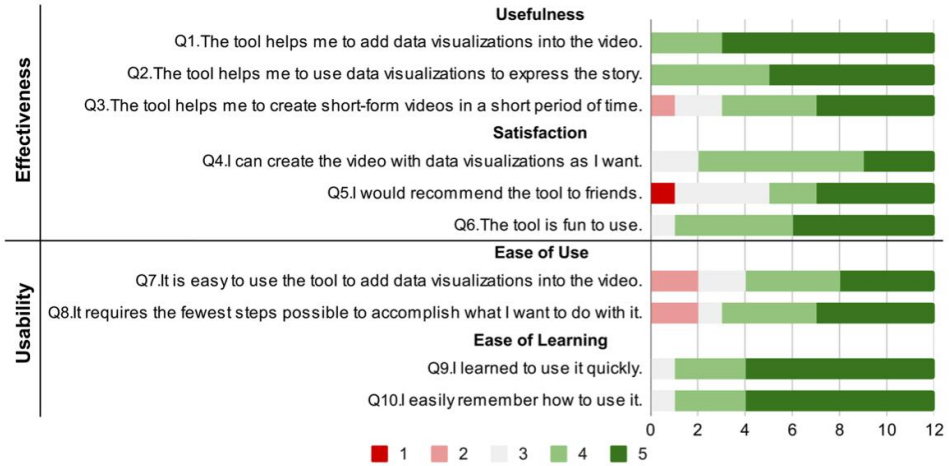
Helping express a story informs perceived usefulness.

LEARNING AND EASE

Learning quickly and using few steps are subjective judgements.

MEMORY CLAIM

A rating of easy remembering does not test actual delayed reuse.



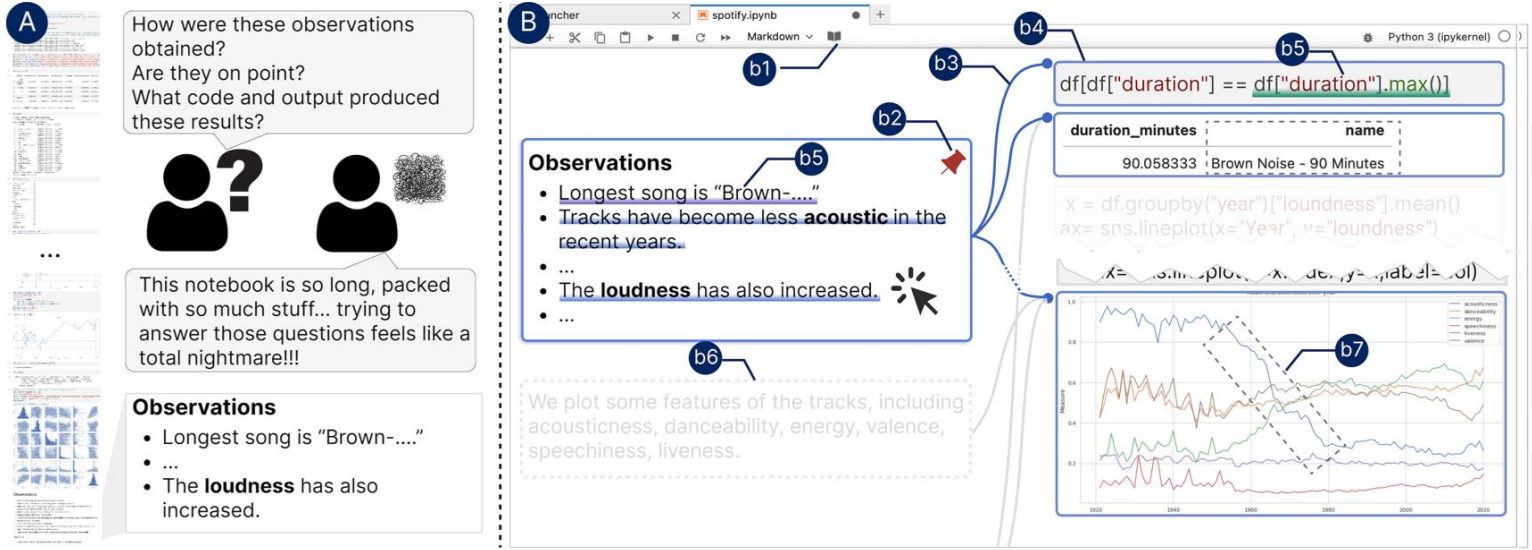
Original Figure 8, This figure shows the questionnaire responses on the effectiveness and usability of VisTellAR. (1:strongly disagree; 5:strongly agree).

Which claim could a return session after a longer gap test?

Tong et al. (2025), VisTellAR, Fig. 8. Lund (2001), USE. Nielsen (2012), Usability 101.

Research Example: InterLink

Lin et al., CHI 2025. Reading explanations, code and charts together



Original Figure 1, InterLink helps analysts read and understand computational notebooks by linking text to code and outputs using a side-by-side layout.

A user needs to trace an explanation to its supporting code or chart.

Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104

DEMO VIDEO

InterLink's Design Process

The paper explicitly connects a formative study, requirements, design and evaluation

OUR STAGE MAPPING	EVIDENCE IN THE PAPER	WHERE TO LOOK
Discover	6 interviews with concrete notebook examples	Section 3.1–3.3
Define	5 design requirements	Section 3.4
Develop	Linked layout and interactions	Section 4
Deliver	12-participant comparison	Section 6

Double Diamond is our teaching mapping. The section references show the authors' documented process.

Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104

InterLink: Evidence to Requirements

A design requirement should answer a documented difficulty

DIFFICULTY	DESIGN REQUIREMENT
Unclear relationships	Show connections without excessive clutter (DR1).
Repeated scrolling	Support an alternative layout (DR3).
Disrupted focus	Allow focus on different content types (DR4).

Three of the five requirements, paraphrased. Paper, Sections 3.3–3.4.

Which requirement explains the need for a layout change?

Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104

InterLink: Experimental Setup

A controlled comparison of two ways to read the same kinds of information

STUDY ELEMENT	REPORTED SETUP
Participants	12 participants
Conditions	InterLink and standard JupyterLab
Materials	Two Kaggle notebooks
Task	8 questions per notebook, requiring cross-cell information
Design	Within-subjects, with counterbalanced interface order

Paper, Section 6.1. The study tests notebook reading tasks.

Which comparison would test your Agent's specific design claim?

Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104
Section 6.1. The task concerns finding and integrating notebook information.

InterLink: What Did They Measure?

Our interpretation connects each reported measure to a design goal

REPORTED MEASURE	WHAT IT CAN INFORM
Correct answers	Usability: task effectiveness
Time per question	Usability: efficiency
System Usability Scale (SUS): 76.5	Overall perceived usability of InterLink
Confidence rating	Self-reported confidence
Inverse Efficiency Score (IES)	Combined performance: time / accuracy

Measures from Section 6. Our goal classification is a teaching interpretation.

Which additional study would test memorability after a gap?

Lin et al. (2025), InterLink, §6. Brooke (1996), SUS.
ISO 9241-11:2018. Goal classifications are our classroom interpretation.

InterLink: Results and Their Meaning

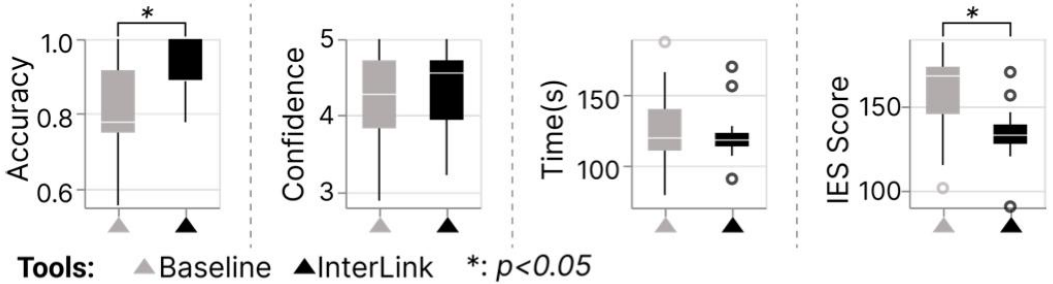
Accuracy improved in these tasks. The time difference was not significant.

TASK ACCURACY

92% versus 81%, with a significant difference ($p < .05$).

TASK TIME

122 versus 128 seconds. The difference was not significant.



CLASS DISCUSSION

Which usability claim has stronger support in this study?

Original Figure 10. Paper, Section 6.2. Baseline is standard JupyterLab.

The conclusion needs the tested users, tasks and comparison.

Lin et al. (2025), InterLink, CHI. DOI: 10.1145/3706598.3714104

Which Construct Did They Measure?

Classify the reported evidence and name a missing test

EVIDENCE	YOUR INTERPRETATION
InterLink: users' correct answers and time	
InterLink: SUS total score	
VisTellAR: "easy to remember" rating	
VisTellAR: perceived task-support rating	
An AI model's answer accuracy	

Which evidence would test learnability, actual memorability and utility in context?

Lin et al. (2025), InterLink. Tong et al. (2025), VisTellAR. Classification exercise adapted for ISA104.

AI Across the Double Diamond

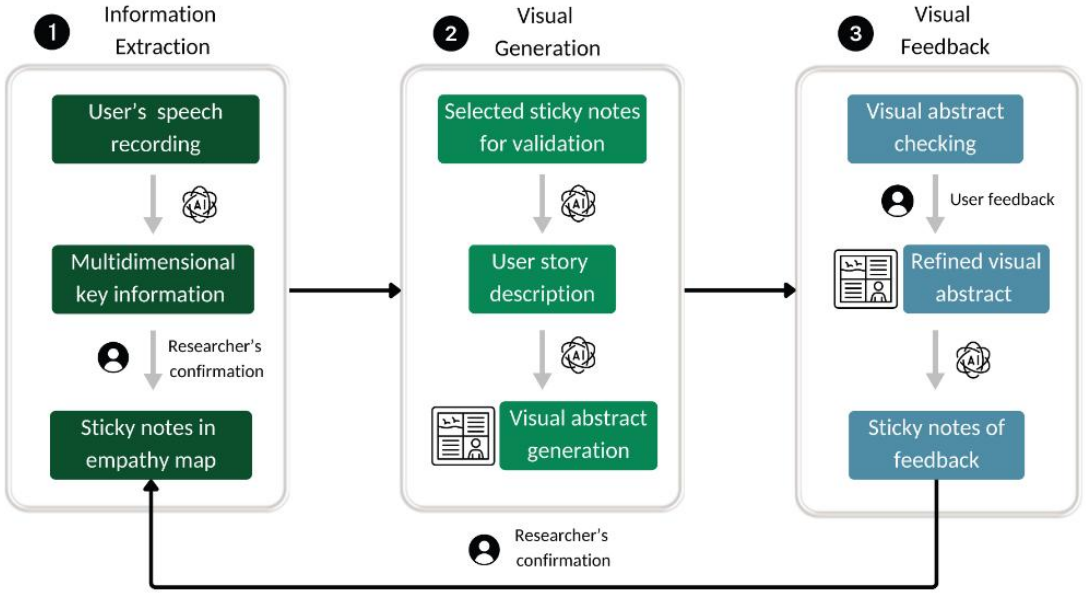
AI can support specific tasks while people retain responsibility for evidence and decisions

STAGE	AI-SUPPORTED ACTIVITY	PAPER EXAMPLE
Discover	Extract interview information and check understanding	InsightBridge, CHI 2025
Define	Suggest text labels and groups for qualitative analysis	CollabCoder, CHI 2024
Develop	Explore and refine alternative concepts	AIdeation, CHI 2025
Deliver	Prepare and rehearse usability evaluation	UXAgent, CHI EA 2025

AI can propose, organise or simulate. People still verify evidence and justify the design decision.

Li et al. (2025), InsightBridge. Gao et al. (2024), CollabCoder.
Wang et al. (2025), AIdeation. Lu et al. (2025), UXAgent.

Discover: AI in User Interviews



Original Figure 1. The workflow of InsightBridge. We leverage LLMs to support the three main functions of InsightBridge, including information extraction, visual generation, and visual feedback. The actions done by human or AI are annotated respectively.

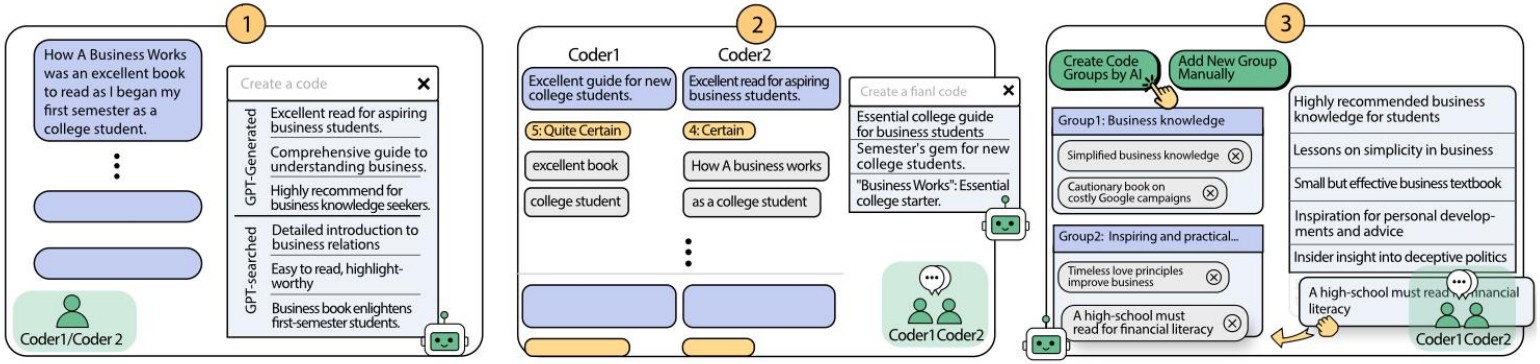
Human check: ask the user whether the AI's representation matches their experience.

Li et al. (2025), InsightBridge, CHI. DOI: 10.1145/3706598.3713640

PAPER VIDEO

Define: AI in Evidence Synthesis

CollabCoder suggests text labels and groups that researchers can inspect and revise



Original Figure 1, CollabCoder, A Lower-barrier, Rigorous Workfow for Inductive Collaborative Qualitative Analysis. The workfow consists of three key stages: 1) Independent Open Coding, facilitated by on-demand code suggestions from LLMs, producing initial codes; 2) Iterative Discussion, focusing on confict mediation within the coding team, producing a list of agreed-upon code decisions; 3) Codebook Development, where code groups may be formed through LLM-generated suggestions, based on the list of decided codes in the previous phase.

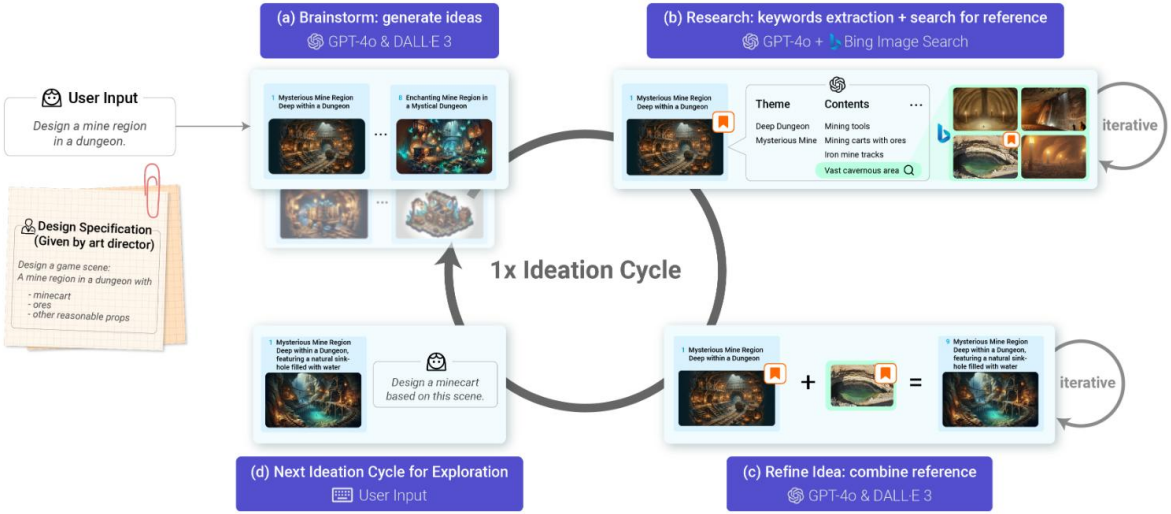
Human check: return to the original text before accepting or merging a theme.

Gao et al. (2024), CollabCoder, CHI. DOI: 10.1145/3613904.3642002

DEMO VIDEO

Develop: AI in Concept Exploration

AIdeation supports brainstorming, reference exploration and iterative recombination



Original Figure 1, AIdeation, an ideation tool designed to support concept designers in exploring ideas in both breadth and depth, with flexible iterative refinement. The figure illustrates the ideation cycle using a real-world project example from our field study, showcasing AIdeation's key design components.

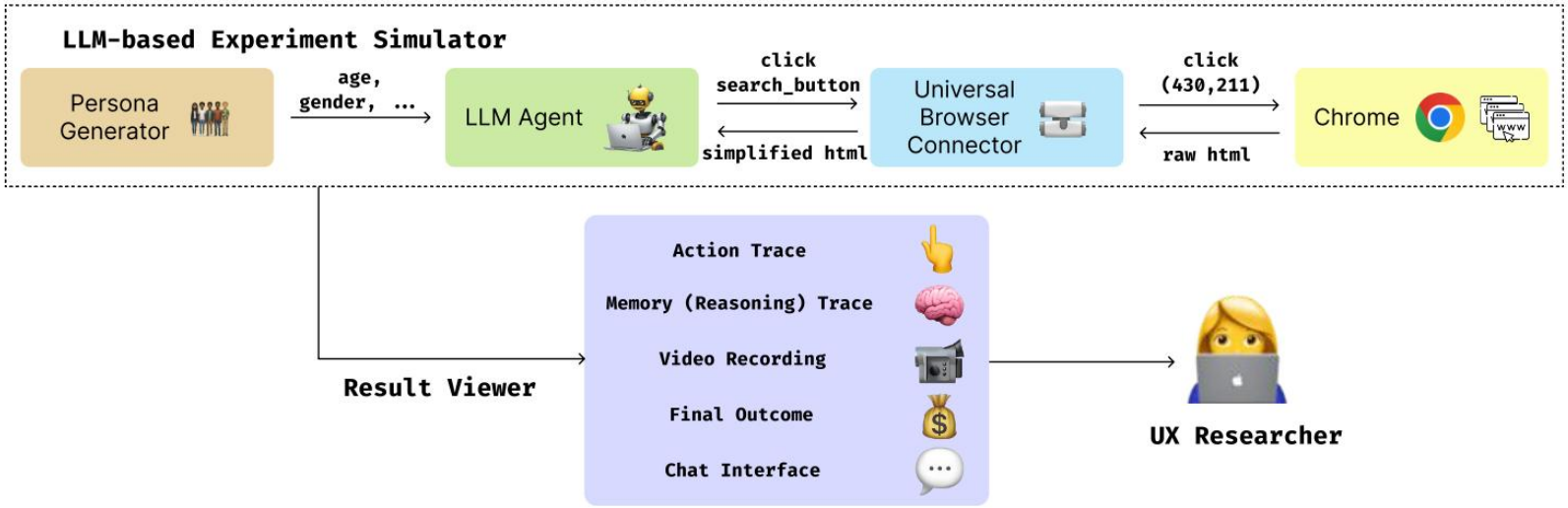
Human check: which alternative best addresses the user's need, and why?

Wang et al. (2025), AIdeation, CHI. DOI: 10.1145/3706598.3714148

PAPER VIDEO

Evaluation: AI for a Test Rehearsal

UXAgent simulates website interactions before a study with real participants



Original Figure 1, UXAgent System Design. An UX researcher can use the persona generator to generate thousands of user personas and feed them into the LLM Agent. The LLM Agent interacts with the target web design through a Universal Browser Connector. Collected multimodal data are presented to the UX researcher through a Result Viewer..

Human check: which findings need confirmation with actual target users?

Lu et al. (2025), UXAgent, CHI Extended Abstracts. DOI: 10.1145/3706599.3719729

PROJECT DEMO

AI Evaluation and Human Evaluation

ProactiveVA evaluates the agent and the people using the system

EVALUATION	REPORTED INDICATORS	MAIN LENS
Agent tasks	Task completion, data accuracy, path efficiency	Technical performance
Human analysis task	Reported events, details and correctness	User task performance
Experience ratings	Clarity, control and disruption	Usability and broader UX

Paper, Sections 5.1 and 5.3. Goal classification is our classroom analysis.

Whose efficiency is measured: the agent's execution or the person's task?

Zhao et al. (2026), ProactiveVA, IEEE TVCG. DOI: 10.1109/TVCG.2025.3642628

PROJECT DEMO

Privacy, Bias and Stereotypes

Two fictional scenarios in an AI-assisted research workflow

PRIVACY LEAKAGE

A transcript contains confidential details that become exposed after upload.

People check permission, minimise identifying details and choose appropriate processing.

BIASES AND STEREOTYPES

An AI summary adds gender-based assumptions absent from the source records.

People compare it with the original evidence and examine whose views are represented.

Ask what information leaves the study and what assumptions the AI adds.

Wu (2024), CMU DHCS, Human–AI Interaction and (Large) Language Models for HCI.

Reproducibility in AI-Assisted Research

Could another researcher understand and repeat the procedure?

WHAT TO RECORD	WHY IT MATTERS
Model, version and access date	The model or service can change.
Prompts, data and conversation context	Small input differences can affect the result.
Settings and tools	The workflow includes more than the final prompt.
Outputs and repeated runs	One response may not represent typical behaviour.

Closed models can limit inspection. Matching settings does not guarantee identical outputs.

Wu (2024), CMU DHCS, (Large) Language Models for HCI. Classroom reporting checklist.

Your Group Agent: Design Evidence

One improvement supported by user evidence, with a plan for AI assistance

STAGE	YOUR EVIDENCE OR DECISION	AI HELP AND HUMAN CHECK
Discover	Who is the user, and what difficulty occurs?	
Define	Which requirement follows from the evidence?	
Develop	Which alternatives and principle guide the design?	
Deliver	Which task and measure will test the claim?	

State one Utility question, two Usability goals and the evidence people will check.

Class activity based on Design Council, The Double Diamond, and Nielsen (2012), Usability 101.

Peer Feedback

Explain one design decision to another group

1 THE NEED	Which evidence supports the problem you chose?
2 THE DESIGN	Which principle explains the proposed change?
3 THE TEST	What dimension does the measure capture, and what will people check?

Give one specific suggestion and revise the plan.

ISA104 group discussion. Design process and evaluation questions adapted from the sources cited above.

Key Takeaways

A design claim needs an evidence chain

THE DESIGN PROCESS	Connect user evidence, a focused need, alternative designs and evaluation.
THE MEASUREMENT	Separate utility, usability and other outcomes. Match questionnaires to constructs.
THE AI ROLE	Use AI for specific design activities and explain the evidence people need to check.

Your next step: one justified improvement to your group Agent.

Design Council, The Double Diamond. Nielsen (2012), Usability 101. Research cases cited throughout.

References: Frameworks and Methods

Sources for the process, usability concepts and research methods

Design Council. The Double Diamond

Four-stage design framework. Original diagram, CC BY 4.0.

Nielsen, J. Usability 101 and 10 Usability Heuristics

NN/g, 2012 and 1994. Also: Norman and Nielsen, The Definition of User Experience.

ISO (2018). ISO 9241-11: Usability, Definitions and Concepts

Ergonomics of human-system interaction, Part 11.

Wu, S. T. (2024). Designing Human-Centered Software

CMU HCII. Supplied lectures on design process and Human–AI/LLM interaction.

References: Questionnaires

Original instruments and their intended constructs

Brooke, J. (1996). SUS: A “Quick and Dirty” Usability Scale

In Usability Evaluation in Industry. Overall subjective usability.

Davis, F. D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology
MIS Quarterly 13(3), 319–340. DOI: 10.2307/249008

Lund, A. M. (2001). Measuring Usability with the USE Questionnaire

Usability Interface 8(2), 3–6. Four categories in the 30-item instrument.

References: Design and Evaluation Cases

Papers used for design processes, experiments and measurement

Lin et al. (2025). InterLink: Linking Text with Code and Output in Computational Notebooks

CHI 2025. DOI: 10.1145/3706598.3714104

Tong et al. (2025). VisTellAR: Embedding Data Visualization to Short-Form Videos Using Mobile Augmented Reality
IEEE TVCG 31(3), 1862–1874. DOI: 10.1109/TVCG.2024.3372104

Zhao et al. (2026). ProactiveVA: Proactive Visual Analytics with LLM-Based UI Agent

IEEE TVCG 32(1), 451–461. DOI: 10.1109/TVCG.2025.3642628

References: AI in Design Activities

Papers illustrating AI assistance at different parts of the process

Li et al. (2025). InsightBridge: Enhancing Empathizing with Users through Real-Time Information Synthesis and Visual Communication
CHI 2025. DOI: 10.1145/3706598.3713640

Gao et al. (2024). CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models
CHI 2024. DOI: 10.1145/3613904.3642002

Wang et al. (2025). AIdeation: Designing a Human-AI Collaborative Ideation System for Concept Designers
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Instructor Appendix: Quiz Coverage 1

Hidden during slideshow. Concepts are taught in the visible slides.

QUIZ	REQUIRED CONCEPT	TEACHING SLIDES
Q1–2	Usability goals and learnability	12, 13, 35
Q3–4	Double Diamond and AI-generated alternatives	3, 4, 9, 36, 39
Q5–6	Field studies, interviews and self-report	5–7, 37
Q7–8	Think-aloud during a task and its purpose	17
Q9–10	Semi-structured interview and focus group	6, 7
Q11–12	Recognition compared with recall	10

Assessment source: supplied SAT101 Quiz, 20 Questions and Answers.

Instructor Appendix: Quiz Coverage 2

Hidden during slideshow. Continue the required-concept map.

QUIZ	REQUIRED CONCEPT	TEACHING SLIDES
Q13–15	Formative improvement and summative assessment	11, 16
Q16	Same participants experience all conditions	18, 32
Q17	Between subjects can reduce carry-over	18, 19
Q18	Confidential information and privacy leakage	42
Q19	Biases and gender stereotypes	42
Q20	Closed LLMs, varying outputs and reproducibility	43

Assessment source: supplied SAT101 Quiz, 20 Questions and Answers.